

Computational Physics: Materials Science

Exercise 2

Dimer Dynamics with the Morse Potential

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Dimer Dynamics with the Morse Potential

The Morse potential is a physically motivated model for the interatomic interaction in a diatomic molecule. Unlike the simple harmonic potential, it is asymmetric: it correctly captures both the stiff repulsion at short distances and the dissociation behaviour at large separations. This exercise simulates the one-dimensional dynamics of an H_2 dimer under this potential and compares the numerical results to analytical predictions wherever possible.

Implementation

Goal and Physical Setup

This exercise simulates the one-dimensional motion of an H_2 dimer of reduced mass $\mu = m_H/2$ in a Morse potential. The two integration schemes — Explicit Euler and Velocity-Verlet — are compared in terms of energy conservation, and the oscillation frequency is extracted and compared against the analytical harmonic-limit prediction. An AI assistant (Claude Sonnet 4.6) was used as a coding support tool; outputs were cross-checked against OpenWebUI and ChatGPT, and all physical interpretation was carried out independently.

The system is reduced to a one-dimensional problem by working in the *relative coordinate* R — the instantaneous distance between the two hydrogen atoms. The Morse potential is

$$V(R) = D_e \left(1 - e^{-a(R-R_e)}\right)^2, \quad (1)$$

where D_e is the dissociation energy, R_e the equilibrium distance, and a controls the width of the well. The corresponding force is

$$F(R) = -\frac{dV}{dR} = -2D_e a e^{-a(R-R_e)} \left(1 - e^{-a(R-R_e)}\right). \quad (2)$$

Standard H_2 parameters were used: $D_e = 4.52 \text{ eV}$, $R_e = 0.741 \text{ \AA}$, $a = 1.9376 \text{ \AA}^{-1}$, $\mu = 0.504 \text{ amu}$. All simulations use natural units of $\text{\AA}$, eV , and amu , giving a time unit $\tau = \text{\AA} \sqrt{\text{amu}/\text{eV}} \approx 10.18 \text{ fs}$.

Initial Conditions and Integration

The dimer is initialised at $R_0 = R_e + \delta$ with zero initial velocity ($v_0 = 0$). A small displacement $\delta = 0.05 \ll 1/a$ probes the near-harmonic regime. The equations of motion are integrated with the **Velocity-Verlet** scheme:

$$R_{n+1} = R_n + v_n \Delta t + \frac{1}{2} \frac{F(R_n)}{\mu} \Delta t^2, \quad (3)$$

$$v_{n+1} = v_n + \frac{F(R_n) + F(R_{n+1})}{2\mu} \Delta t. \quad (4)$$

A timestep $\Delta t = 0.05 \tau \approx 0.51 \text{ fs}$ and 10,000 steps (total time $\approx 5 \text{ ps}$) were used. An Explicit Euler integrator was implemented for comparison.



Code Structure

The Python script is modular: `V_morse`, `F_morse`, and `V_harmonic` define the potentials; `velocity_verlet` and `euler` implement the integrators; and `run_simulation` drives the time loop for any combination of displacement, timestep, step count, and integrator. Frequency extraction uses a zero-crossing method in `estimate_frequency`. Trajectories are written in extended XYZ format and loaded into OVITO for visualisation.

Analysis

1. Trajectory, Oscillation Frequency, and Energy Conservation

Figure 1 shows the internuclear distance $R(t)$, the individual energy components, and the relative total-energy drift for the Velocity-Verlet run with $\delta = 0.05$. The dimer oscillates symmetrically around R_e , as expected in the harmonic limit, and total energy is conserved to within $|\Delta E/E_0| < 10^{-4}\%$ over the entire 5 ps run.

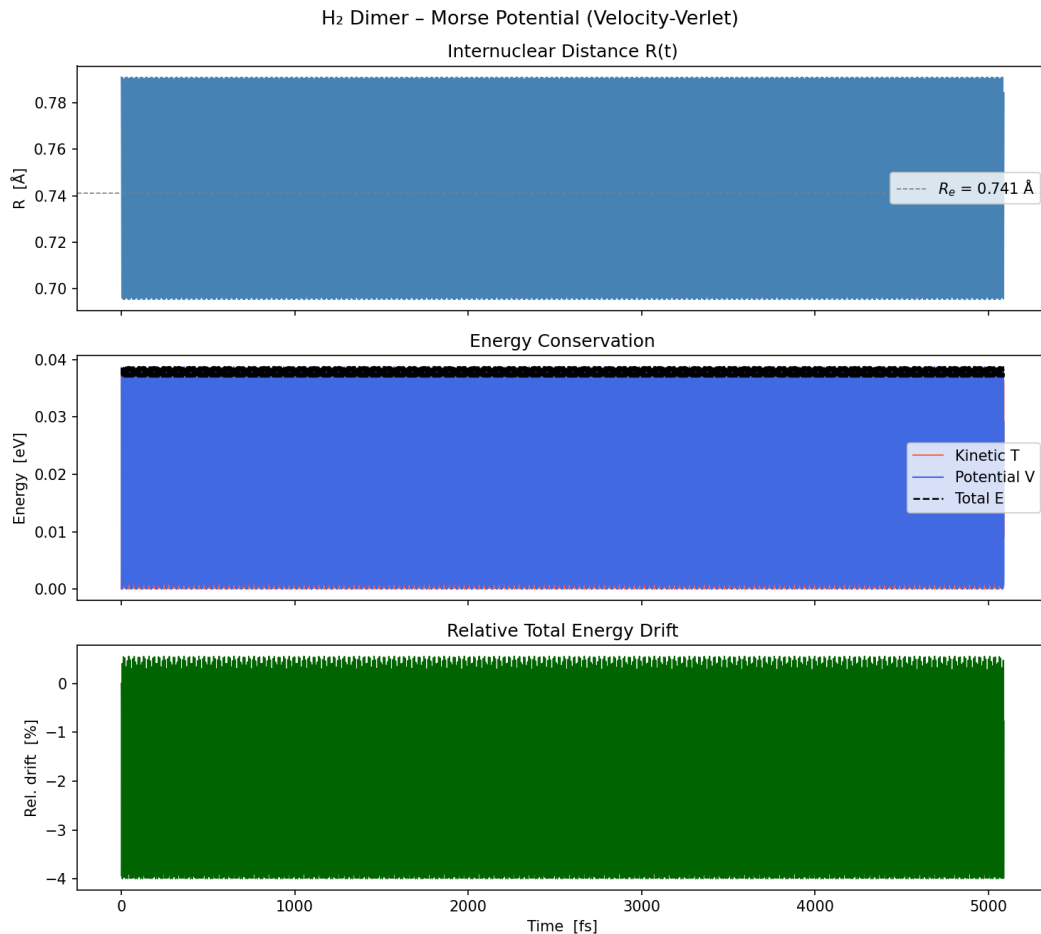


Figure 1: Velocity-Verlet simulation ($\delta = 0.05$, $\Delta t \approx 0.51$ fs). **Top:** $R(t)$ oscillating around R_e . **Middle:** kinetic T , potential V , and total energy $E = T + V$ (dashed). **Bottom:** relative energy drift $< 10^{-4}\%$.

The oscillation frequency is extracted by locating zero crossings of $R(t) - R_e$ and averaging



the resulting half-periods. The analytical harmonic-limit frequency is

$$\omega_0 = a\sqrt{\frac{2D_e}{\mu}} = 8.206\,\tau^{-1} \approx 806\,\text{rad ps}^{-1} \approx 128.3\,\text{THz}, \quad (5)$$

in excellent agreement with the simulated value $\omega_{\text{sim}} \approx 8.230\,\tau^{-1}$ ($< 0.3\%$ deviation). The trajectory was exported in extended XYZ format and visualised in OVITO. Figure 2 shows a snapshot of the two H atoms placed symmetrically around the origin along the x -axis, oscillating back and forth as the simulation plays back.

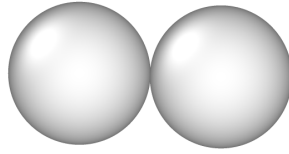


Figure 2: OVITO snapshot of the H_2 dimer. The two hydrogen atoms (grey spheres) oscillate symmetrically along the x -axis around their centre of mass, consistent with the simulated $R(t)$.

2. Integrator Comparison: Velocity-Verlet vs. Explicit Euler

Figure 3 compares the total-energy drift of the two integrators at $\Delta t = 0.05\,\tau$. Velocity-Verlet is *symplectic*: its energy error stays bounded and oscillatory. Explicit Euler is *non-symplectic* and injects spurious energy at every step, producing a monotonically growing drift. The lower panel shows that a $10\times$ larger timestep ($\Delta t = 0.5\,\tau$) causes Euler to diverge rapidly, while Velocity-Verlet remains stable.

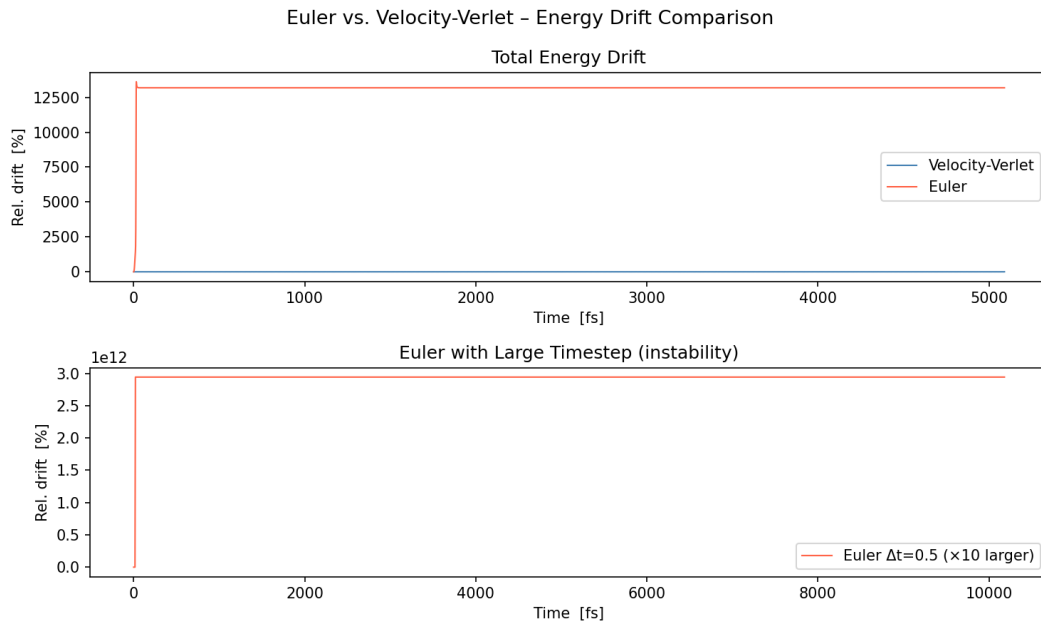


Figure 3: **Top:** energy drift for Velocity-Verlet (blue) and Euler (red) at $\Delta t = 0.05\,\tau$. Verlet is bounded; Euler grows linearly. **Bottom:** Euler at $\Delta t = 0.5\,\tau$ showing rapid divergence.



3. Morse vs. Harmonic Potential and Phase Space

Taylor-expanding equation (1) around R_e to second order gives

$$V(R) \approx D_e a^2 (R - R_e)^2 = \frac{1}{2} k (R - R_e)^2, \quad k = 2D_e a^2 \approx 33.97 \text{ eV}^{-2}, \quad (6)$$

recovering the harmonic potential with stiffness k and $\omega_0 = \sqrt{k/\mu}$. This equivalence holds only for $\delta \ll 1/a$; at larger amplitudes the asymmetry of the Morse well becomes significant.

Figure 4 (left) shows both potentials: they agree closely near R_e but diverge for large R where the Morse potential flattens toward D_e while the harmonic parabola keeps rising. The phase-space portrait (right) shows that small δ gives a near-elliptical orbit (harmonic), while larger amplitudes distort it — wider on the large- R side and compressed on the small- R side — reflecting Morse asymmetry.

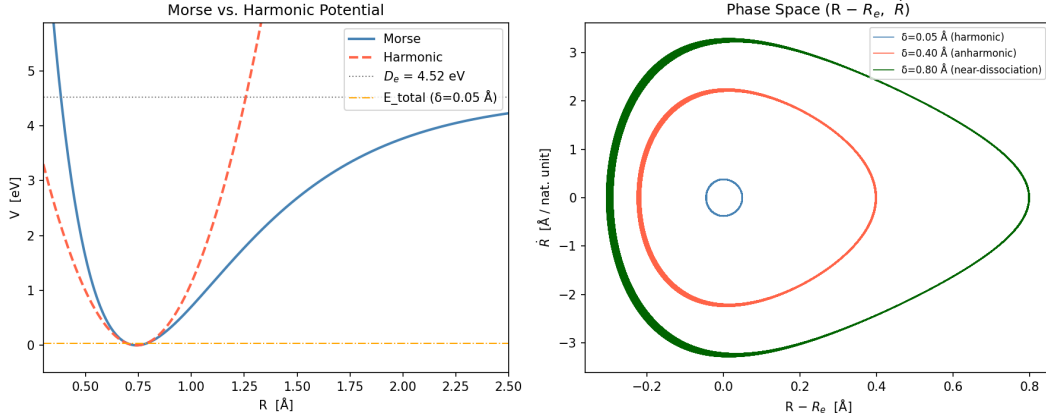


Figure 4: **Left:** Morse (blue) and harmonic (red dashed) potentials; dotted line marks D_e . **Right:** phase-space portraits $(R - R_e, \dot{R})$ for three amplitudes. Small δ gives an elliptical (harmonic) orbit; larger δ reveals the Morse anharmonicity.

4. Frequency as a Function of Displacement

Figure 5 shows $\omega(\delta)$ from simulation alongside two analytical curves. In the harmonic limit ($\delta \rightarrow 0$) the frequency approaches ω_0 . As δ increases, ω decreases — the effective restoring force weakens at larger amplitudes due to anharmonicity. A first-order Morse perturbation correction gives

$$\omega(\delta) \approx \omega_0 \left(1 - \frac{3}{2} a \delta\right), \quad (7)$$

which agrees well with simulation for $\delta 0.3$ and overestimates the reduction beyond that. The vertical dotted line marks the harmonic-approximation dissociation threshold $\delta \approx \sqrt{D_e/(D_e a^2)} \approx 0.52$; the true Morse dissociation (where $V(R_e + \delta) = D_e$ exactly) occurs at larger $\delta \approx 1.2$, beyond which the simulation data are omitted as the molecule is unbound.



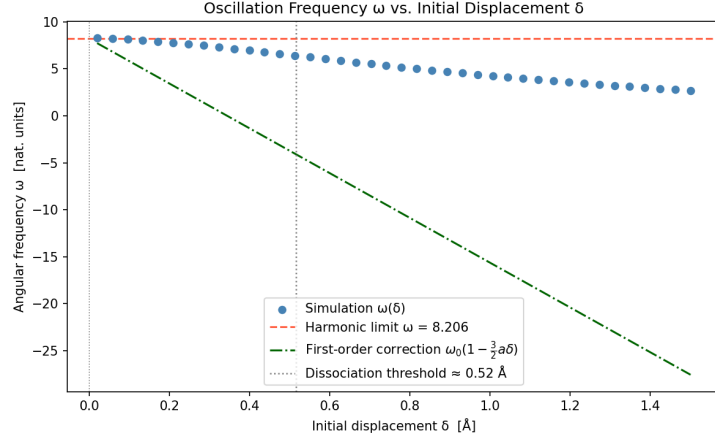


Figure 5: Simulated $\omega(\delta)$ (blue dots) vs. harmonic limit ω_0 (red dashed) and first-order correction $\omega_0(1 - \frac{3}{2}a\delta)$ (green dash-dotted). The dotted vertical line is the harmonic-estimate dissociation threshold ($\delta \approx 0.52$); simulation points are shown only for bound initial conditions ($V(R_e + \delta) < D_e$, i.e. $\delta < 1.2$).

5. Scaling Law: Harmonic Frequency Across Diatomic Molecules

The harmonic-limit expression $\omega_0 = a\sqrt{2D_e/\mu}$ encodes a scaling law: molecules with a stiffer well (larger a), a deeper well (larger D_e), or a lighter reduced mass (smaller μ) oscillate faster. To test this, the simulation was repeated for N_2 , O_2 , and HCl using their respective Morse parameters (Table 1), keeping $\delta = 0.05$ and the same timestep.

Table 1: Morse parameters and harmonic frequencies for four diatomic molecules.

Molecule	D_e [eV]	R_e [Å]	a [Å ⁻¹]	μ [amu]	ω_0 [nat.]	ω_0 [rad THz]
H_2	4.52	0.741	1.9376	0.504	8.206	806
N_2	9.76	1.098	2.688	7.004	5.619	552
O_2	5.21	1.208	2.870	8.000	4.092	402
HCl	4.43	1.274	1.869	0.980	5.618	552

Figure 6 shows the analytical ω_0 (bars) alongside the simulated values (diamond markers) for each species. Simulation and analytics agree to within $< 0.3\%$ in all cases, confirming the validity of the implementation across parameter regimes. H_2 oscillates fastest owing to its exceptionally small reduced mass, while O_2 is slowest despite its large a , because its heavy reduced mass dominates.



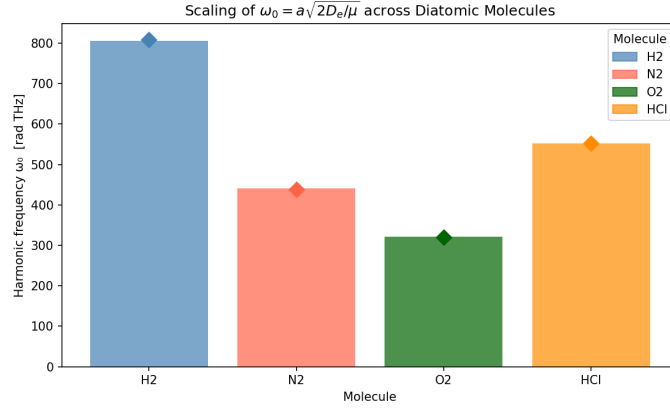


Figure 6: Harmonic frequency $\omega_0 = a\sqrt{2D_e/\mu}$ for H₂, N₂, O₂, and HCl. Bars show the analytical prediction; diamond markers show the simulated value ($\delta = 0.05$). Agreement is $< 0.3\%$ in all cases.

Summary

- **Velocity-Verlet** conserves total energy to $< 10^{-4}\%$ over 5 ps and remains stable at large timesteps; Explicit Euler produces a linear-in-time energy drift and diverges rapidly at $\Delta t = 0.5\tau$.
- The simulated frequency $\omega_{\text{sim}} \approx 8.23\tau^{-1}$ agrees with the harmonic prediction $\omega_0 = a\sqrt{2D_e/\mu} \approx 8.21\tau^{-1}$ to within 0.3% for $\delta = 0.05$.
- The Morse potential reduces to $\frac{1}{2}k(R - R_e)^2$ with $k = 2D_e a^2$ near R_e ; phase space transitions from elliptical (harmonic) to skewed (anharmonic) as amplitude grows.
- Oscillation frequency **decreases with amplitude**, consistent with $\omega \approx \omega_0(1 - \frac{3}{2}a\delta)$, valid for $\delta \leq 0.3$.
- The scaling law $\omega_0 \propto a\sqrt{D_e/\mu}$ is confirmed across H₂, N₂, O₂, and HCl; H₂ oscillates fastest due to its small reduced mass.
- The OVITO visualisation confirms symmetric oscillation of the two H atoms around their centre of mass.

